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Effectiveness of interventions on healthcare professionals' understanding and use of conscience: A systematic review protocol.

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Manuscripts

Title: Effectiveness of interventions on healthcare professionals' understanding and use of conscience: A systematic review protocol.

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Abstract

Introduction: Conscience is central to moral decision making. In the context of morally pluralistic workplaces today, healthcare professionals' conscience may prompt them to make moral decisions to refrain from providing services they morally disagree with. However, such decisions are largely viewed as in the literature as contentious, giving rise to polarizing arguments for and against healthcare professionals' freedom of conscience. Yet, little work has been done to understand and support healthcare professionals' conscience. Therefore, there is limited research over the degree and extent to which healthcare professionals are supported to understand and use their conscience. The objective of this review is to examine the scholarly evidence to ascertain which interventions do exist, and to examine how effective they are in supporting healthcare professionals' understanding and use of conscience in practice.

Methods and Analyses: Independent reviewers will systematically review ten inter-disciplinary, scholarly databases to examine qualitative, quantitative and mixed-methods studies including clinical trials pertaining to interventions related to conscience for healthcare professionals. Databases to be searched include: the Cochrane Controlled Register of Trials, Medline, EMBASE, PsycINFO, CINAHL, Academic Search Complete, ATLA Religion Database, Religion and Philosophy Collection, PhilPapers and Scopus for studies in English from 2000 onwards. Databases were searched in May, 2021. Study screening, selection, extraction and risk of bias assessments on each study using the Mixed Methods Appraisal Tool will be independently conducted by reviewers. Descriptive data synthesis will be carried out. Statistical analysis and meta-analysis will be conducted as relevant, based on homogeneity of findings. The quality of the aggregate evidence will be assessed using the Grading of Recommendations Assessment, Development and Evaluation Tool criteria.

Ethics and dissemination: Ethical approval is not required for this review. This protocol will not involve patients or patient information. The results will be reported in a peer-reviewed journal and disseminated at conferences.

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Strengths and limitations (5 bullet point sentences, related to the methods)

- This study aims to examine interventions used to enhance HCPs' understanding and use of conscience.
- Study selection, data extraction and risk of bias assessments will be done independently by a minimum of two reviewers.
- Studies included in this review will be limited to English language articles, which may lead to a risk for publication bias.
- The strength of evidence in this study will be evaluated according to the Grading of Recommendations Assessment, Development and Evaluation criteria.
- Inclusion of various methodological perspectives and methods might capture more interventions.

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Introduction

Conscience is a fundamental human freedom and an essential concept for bioethics and the health sciences.[1,2,3] Conscience is the ability to make decisions based on information derived from moral knowledge in order to choose to do something that is ethical or refrain from doing something unethical.[4,5] As such, conscience is central to ethical healthcare in which healthcare professionals (HCPs) are required to make moral decisions to engage in practice that is ethical and beneficial for their patients.

However, in bioethics and the health sciences today, little attention has been paid to the meaning of conscience as it pertains to healthcare professionals' (HCP) moral decision making. Instead, issues around conscience, such as conscientious objection, are becoming increasingly polarized in today's morally pluralistic societies and healthcare communities owing to the lack of agreement that may occur in pluralistic contexts. Subsequently, disagreement on how HCPs express their issues of conscience is also increasing owing to the lack of understanding regarding what conscience is in the first place.[6] This knowledge gap is problematic because the scholarly literature is fraught with academic dispute over how to handle issues of conscience. Amidst this debate, little attention is being given to HCPs with respect to expressing their issues of conscience and what conscience and issues of conscience mean to them. For the purpose of this review, HCPs consist of physicians, nurses and midwives who have all been documented in the scholarly literature as having to make ethical decisions in the course of their practice which involves the use of their conscience.[7,8,9] Additionally, international organizations such as the World Medical Association, the International Council of Nurses and the International Confederation of Midwives stipulate that conscience and the human right to freedom of conscience are relevant to medicine, nursing and midwifery.[3,2,10]

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The research that has been done on the meaning of conscience for HCPs indicates that physicians, nurses and midwives alike voice that conscience is important to their ability to make moral meaning of their ethical practice.[7,8,11] Importantly, conscience is well taken up in the philosophical and theological literature. Within this scholarly space, conscience has a long-standing conceptual history.[11,12] In this body of literature, conscience has emerged as a necessary component for being a moral person.[13,14] More recently, empirical work in moral psychology is revealing compelling empirical evidence that morality is central to human behaviour.[15-17] Philosophy and theology are also taken up in bioethics, which is an inter-disciplinary discipline devoted to asking essential life questions that support healthcare providers and patients to make ethical decisions in the context of healthcare.[18,19] Owing to the traction that conscience has in these inter-disciplinary disciplines, we will include inter-disciplinary databases in our search to ascertain if there are interventions in the inter-disciplinary as well as the healthcare and bioethics literature related to our research questions. Doing so may show how an integrated understanding of conscience in philosophy and theology can inform HCP's understanding and use of conscience for ethical healthcare education and practice today. Specifically, by supporting HCPs to use their conscience and increasing an awareness of the significance that conscience holds for ethical decision making. Doing so might start to address the gap related to understanding conscience in healthcare communities and may offer a morally inclusive way to support HCPs' moral decision making in practice. Advancing more understanding in relation to conscience and the role it holds in the human experience across healthcare contexts has the potential for global impact given that freedom of conscience is a universal, human right. This review might generate

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further empirical ethical research to benefit HCP's ethical practice and the patient populations they care for.

To start to address this knowledge gap related to ethics around respect for HCPs' understanding and use of conscience, we will conduct an up-to-date systematic review of the interdisciplinary, scholarly literature on interventions utilized to support healthcare professionals' understanding and use of conscience. To the best of our knowledge, no systematic reviews have been undertaken in relation to this gap in ethical knowledge. Disciplines that will be considered are: nursing; medicine; midwifery; bioethics; philosophy; theology; religious studies and moral psychology. In conducting this systematic review, the authors seek to answer the following research questions:

1. What interventions exist in the scholarly, inter-disciplinary literature aimed at enhancing healthcare professionals' understanding and use of conscience?
2. How effective have existing interventions been at supporting health care professionals' use and understanding of conscience?

To answer the research questions the primary objectives of this review study are:

1. To review the evidence about the interventions that were utilized.
2. To determine whether the interventions were effective in providing or improving HCPs understanding and use of conscience.
3. To explore the effectiveness of these interventions as they pertain to the physician, nursing and midwife subgroups and to assess if there is any difference in the effectiveness of the interventions amongst them.

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Methods

Research design and methodology

We will conduct a systematic review (SR) of the published, scholarly, inter-disciplinary literature to examine and explore the effectiveness of interventions utilized to support HCP's understanding and use of conscience in healthcare practice. To initiate the SR study we will follow this review protocol. This protocol will be registered in PROSPERO. The review will be guided by the Preferred Reporting Items for Systematic Review and Meta-analysis 2020 Statement for standardized reporting guidelines, the PRISMA extension checklist for searches (PRISMA-S) and the PRISMA 2020 statement for writing abstracts.[20,21,22] The PRISMA-S checklist will be followed to conduct and document the literature search.[21] This protocol was generated following the Preferred Reporting Item for Systematic review and Meta-Analysis Protocols (PRISMA-P) checklist.[22] (See Research Checklist).

Eligibility criteria

Inclusion criteria

To isolate eligible English-language studies, we will follow the Population, Intervention, Comparators, Outcomes and Study design (PICOS) framework as stipulated in the PRISMA (2020) statement as follows:

Population: Healthcare professionals who engage in clinical practice as identified across the sub-groups of physicians, nurses, and midwives. **Interventions:** Interventions that will be examined are those that aim to understand and address HCP's use of conscience. Possible interventions are: experiments, education and quality improvement initiatives, surveys, interviews, focus groups, experiential inquiries and narrative studies as well as studies that include trials, surveys and pre-post-test data sets. *Sample interventions:* studies and papers to be

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included are those that use primary data and aim to support, enhance, show and address the use and understanding and use of conscience for HCPs. This may occur through terms closely related to conscience such as conscientious objection. Specifically, studies involving the Stress of Conscience Questionnaire and the Perception of Conscience Questionnaire as part of an intervention will be included since they directly relate to the research questions of this SR.[23, 24]

Comparators: no support or opposition to supporting HCPs to understand and use their conscience in clinical practice. **Outcomes:** the types of outcomes that will be sought from the data are the: (1) findings related to what interventions support HCPs' to understand and use their conscience; (2) reports on how HCPs felt supported to use and / or understand their conscience; (3) effects of HCPs being able to voice their issues of conscience in healthcare practice across the sub-groups and (4) what effects that support for conscience has on HCPs across these sub-groups in relation to the quality of care they provide to patients, in relation to themselves, working with colleagues and in their professions. **Study designs:** studies involving interventions may include experimental, quasi-experimental and non-experimental studies consisting of clinical trials, qualitative, quantitative and mixed-method studies.

Patient and Public Involvement: no patients involved.

Exclusion criteria

Articles pre-dating 2020 are ineligible owing to the general lack of empirical research related to conscience prior to this date. Unpublished articles including theses and dissertations will not be included. Papers that utilize secondary data sets such as protocols and reviews are also ineligible.

Search Strategy:

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This review will be reported in adherence to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement with methodological guidance from the Cochrane Handbook of Systematic Reviews of Interventions.[20,25] Further, a systematic search will be developed and conducted by an experienced health sciences librarian (MK) in adherence to the PRISMA for Searching (PRISMA-S) extension.[21]

A preliminary search using the database, Medline, was conducted in April 2021 by MK to determine the feasibility of the review and inform the systematic search strategy (see Table 1 for full search strategy). The search strategy was derived from three main concepts: 1) conscience; 2) healthcare professionals, physicians, nurses, and midwives; 3) interventions, assessment, and quality improvement. Systematic, scoping, and other types of synthesis reviews and their protocols were removed from the results. Finally, results were limited to English language studies published from the year 2000 to present.

(Insert Table 1 here:)

Table 1: **Search Strategy for Medline(R) ALL via OVID (1946 - April 26, 2021).**
Date of search: 27 April 2021

1	conscience/	1574
2	(conscience? or "conscientious* object*").mp.	3707
3	"empirical ethics".mp.	151
4	1 or 2 or 3	3857
5	health personnel/ or exp nurses/ or exp nurse practitioners/ or exp nurse specialists/ or exp nurses, pediatric/ or exp physicians/	278364
6	(doctor* or physician* or "general practitioner*" or GP or surgeon* or anesthesiologist* or cardiologist* or dermatologist* or radiologist* or internist* or geneticist* or neurologist* or obstetrician* or gyn?ecologist* OBGYN or OB-GYN or ophthalmologist* or pathologist* or p?ediatrician* or psychiatrist* or oncologist* or urologist* or allergist* or endocrinologist* or gastroenterologist* or geriatrician* or hospitalist* or nephrologist* or pulmonologist* or rheumatologist* or clinician* or nurse* or midwife* or midwives or ((healthcare or "health care" or health or medical or nursing) adj3 (professional* or personnel or practitioner* or provider*))).mp.	1905235
7	5 or 6	1907349
8	(intervention* or train* or educat* or workshop* or seminar* or curricul* or coach* or mentor* or experiment* or tool* or interview* or "focus group" or review* or screening or	

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questionnaire* or survey* or perception* or implication* or barrier* or facilitat* or obstacle* or encourag* or impediment* or impede* or challeng* or obstruct* or hurdle or experience* or perceive* or perspective* or perception* or "self-report" or trial* or test* or "quality improvement*" or "quality assurance" or "patient safety").mp. 14879880
9 ((QI or "quality improvement*") adj6 (intervention* or initiative* or strateg* or program* or campaign* or implement*)).mp. 18223
10 ((Practice or behavio* or organizational) adj3 chang*).mp. 88479
11 education/ or curriculum/ or education, distance/ or exp education, professional/ or exp education, continuing/ or exp education, graduate/ or exp education, medical/ or exp education, nursing/ or exp inservice training/ or mentoring/ 368239
12 focus groups/ or interviews as topic/ or "surveys and questionnaires"/ or health care surveys/ or self report/ 617218
13 Quality Improvement/27360
14 (((research or knowledge or evidence) adj2 (uptake or "use" or diffus$ or disseminat$ or utiliz$ or utilis$ or transfer$ or translat$ or implement$ or adopt$)) or (innovation$ adj2 adopt$) or (innovation$ adj2 disseminat$) or "research into practice" or "evidence into practice" or "knowledge to action" or "know do gap" or (knowledge adj (mobilization or exchange)) or "translational science" or "implementation science").ti,ab. 70172
15 exp "Diffusion of Innovation"/ or Organizational Innovation/ 43888
16 (evidence-informed adj (healthcare or health care or decision making)).ti,ab. 241
17 (guideline* adj2 (introduc*or issu* or impact or effect* or distribut* or adher* or compl* or utiliz* or utilis* or "use" or uptake or diffuse* or transfer* or implement* or translat* or disseminat* or adopt*)).ti,ab. 20497
18 exp *Evidence-Based Practice/ or Information dissemination/ or Knowledge management/ or Health Knowledge, Attitudes, Practice/ 166249
19 or/8-1814952921
20 4 and 7 and 19893
21 ("systematic* review" or "scoping review" or "synthesis review" or "realist review" or "integrative review" or "meta-anal*" or "metanal* meta-synths?s" or metasynthes?s or meta-narrative or metanarrative or "umbrella review" or hta or "health technology assessment").ti. or systematic review/ or meta-analysis.pt. or technology assessment, biomedical/ or biomedical technology assessment/ or review.pt. 2950266
22 20 not 21 762
23 limit 22 to (english language and yr="2000 -Current") 510

In order to enhance the quality of the systematic search, the preliminary search strategy for Medline was evaluated by an external health sciences librarian using Peer Review of Electronic Search Strategies (PRESS) guidelines.[26] No major revisions were required following the PRESS evaluation but the following recommendations were made: 1) strategy for capturing synthesis reviews was refined to make it more precise for exclusion purposes; 2) additional

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terminology related to knowledge translation were added to the intervention concept (see Supplementary File). The search strategy was revised to include these recommendations and in total, 510 articles were identified from the preliminary search. Out of these results, 30 were deemed relevant for inclusion and confirmed the feasibility of this review.

To identify all relevant published studies, the following databases will be searched:

- Medline via OVID (1946 - Present)
- EMBASE via OVID (1974 - Present)
- PsycINFO via OVID (1806 - Present)
- Cumulative Index for Nursing and Allied Health Literature (CINAHL) via EBSCOhost (1936 - Present)
- Academic Search Complete via EBSCOhost (1887 - Present)
- ATLA Religion Database via EBSCOhost (13th C. - Present)
- Religion and Philosophy Collection via EBSCOhost (1911 - Present)
- PhilPapers via Philosophical Research Online (2001 - Present)
- Scopus via Elsevier (1976 - Present)
- Cochrane Controlled Register of Trials (CENTRAL) via Wiley (1993 - Present)

These databases will be searched using a combination of natural language vocabulary (keywords) and controlled terms (subject headings) wherever they are available, based on the previously identified search concepts. Records identified through database searches will be exported in complete batches and duplicates will automatically be removed upon import to the systematic review management software, Covidence.[27] An update of the searches is planned for six months after search results are exported or one month prior to manuscript submission for publication, whichever is sooner. There is no patient or public involvement in this study.

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Screening

Articles will be screened by multiple reviewers related to the eligibility criteria. Both title and abstract and full-text screening will be done through Covidence software. To facilitate the screening process, a minimum of two reviewers will independently screen all records by title and abstract and retain articles that meet the eligibility and screening question criteria. Included articles for full-text review will again be screened in full by a minimum of two, independent reviewers. Prior to each step in the screening process, reviewers will initially screen approximately ten records and meet to discuss the process to flesh out any questions or disagreements and before carrying out the full screening process. Reviewers will meet throughout each screening process to discuss progress and any conflicts that arise. Throughout the screening and data extraction processes the PI will be responsible for resolving disagreements if consensus is not reached amongst the researchers. Full articles that meet the eligibility criteria will be extracted for data corresponding to variables pre-determined by the research team. The screening and selection process will be shown in a PRISMA flow chart.

Data extraction

Data from included, full text articles will be extracted by a minimum of two, independent reviewers according to pre-set headings developed by the research team. Data headings will be comprised of the following variables in relation to the research questions, PICOS criteria as well as primary and secondary objectives: full reference of the articles; region and country the study was conducted in; research objectives; study design and methods (sample size, comparison types); population under study including sub-groups; description of the intervention and control interventions (where applicable); effects of the intervention (whether the study supported HCPs

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to understand and use their conscience and how, as well as to what extent) in relation to the study outcomes listed in the PICOS criteria which will be captured as significant findings in terms of outcomes, effects of outcomes and limitations. Extracted data will be entered into Word or Excel documents under the data extraction headings to be tabulated for ease of data synthesis and reporting.

Critical appraisal: Assessing risk of bias

Since mixed-methods studies will be included in this SR, extracted data will be critically appraised for methodological quality using the Mixed Methods Assessment Tool (MMAT).[28] At least two, independent researchers will perform this assessment in each individual study. The MMAT is a critical appraisal tool aimed at assessing the quality of studies included in systematic reviews. All studies are appraised against 5 categories of criteria for validity and rigour in relation to study designs comprised of either qualitative, quantitative (randomized trials, non-randomized and descriptive studies) and mixed-methods papers. Reviewers capture their assessments according to core criteria across study designs by either responding 'yes', 'no' or 'can't tell'. Two, independent reviewers will appraise each study using the MMAT and resolve any discrepancies with a third reviewer or the PI. Assessments for each study will be tabulated and presented for data synthesis.

Data synthesis

It is expected that the studies identified will have high methodological heterogeneity – featuring a mix of qualitative, quantitative and mixed-method studies. Accordingly, extracted data will be synthesized descriptively according to relevance of interventions by effect (whether they aimed to support HCPs in understanding and /or using their conscience). The data will be discussed in text and tables to summarize and describe the characteristics and relationship of

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findings across the included studies.[22] Findings will be reported according to fixed categories including: the study year, purpose, design and setting, population, intervention and effect, primary outcome and limitations. Since this SR will include non-quantitative studies, the heterogeneity of the findings will determine whether there is sufficient data to conduct statistical meta-analysis via pooling of outcomes data from sufficiently similar intervention studies. If this is possible, given the wide likely variability across studies, a random effects model will be used for the meta-analysis. If statistical analysis is not applicable, all data will be reported through a narrative descriptive analysis. Quality of the included studies will be assessed using the appropriate checklist tools from the Critical Appraisal Skills Program (CASP); with mixed method studies assessed using the Mixed Method Appraisal Tool (MMAT) for mixed-method studies.[29, 28]

Confidence in cumulative evidence assessment

The Grading of Recommendations Assessment, Development and Evaluation (GRADE) tool will be used to evaluate the strength of the evidence of the included studies.[30]

Quality Assurance

The proposed SR study will follow the PRISMA (2020) format for reporting systematic reviews.

Discussion

This study will review current, inter-disciplinary research studies explicating interventions that were used to understand and / or support HCP's use of conscience. The results of this systematic review will be useful in identifying what concrete supports exist in relation to moral inclusion of HCP's freedom of conscience. Additional gaps may be identified leading to

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further empirical ethics research necessary to support HCP populations to navigate today's morally complex healthcare context.

Acknowledgements: We thank the second research librarian for their PRESS review of the initial Medline search.

Ethics and dissemination: Ethics approval is not required for this study. Findings will be disseminated through publication in a peer-reviewed journal and pertinent conferences.

Author's contributions: CL conceptualized and drafted the protocol. MK and CL created the search strategy. MK conducted the initial Medline search, had the search strategy peer-reviewed, and searched the databases. AMC provided statistical support. AMC, EP and KK contributed to the development of the search criteria and conceptualization of the protocol. KK contributed to the assessment of ROB tools. YBM contributed to the assessment of the strengths and limitations of the research methods. All authors read, provided feedback or approved the manuscript for submission.

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Disclaimer: The work undertaken in this research and all the content and conclusions therein are that of the authors and not the funders.

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Competing interests statement: The PI has funds from the grants listed in the Funding Statement to support open access publication fees for this manuscript if it is accepted for publication.

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I am one author signing on behalf of all co-authors of the Work.

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PRESS Guideline — Search Submission & Peer Review Assessment

SEARCH SUBMISSION: THIS SECTION TO BE FILLED IN BY THE SEARCHER

Searcher: MK
Email: Will be provided after peer review for the purpose of anonymous peer-review of the protocol manuscript.
Date submitted: April 20, 2021
Date requested by: TBD.
[Maximum = 5 working days]

Systematic Review Title:
Effectiveness of interventions on healthcare professionals' understanding and use of conscience: A systematic review protocol.

This search strategy is …

X	My PRIMARY (core) database strategy — First time submitting a strategy for search question and database
	My PRIMARY (core) strategy — Follow-up review NOT the first time submitting a strategy for search question and database. If this is a response to peer review, itemize the changes made to the review suggestions
	SECONDARY search strategy— First time submitting a strategy for search question and database
	SECONDARY search strategy — NOT the first time submitting a strategy for search question and database. If this is a response to peer review, itemize the changes made to the review suggestions

Database
(i.e., MEDLINE,CINAHL…): [mandatory]
MEDLINE

Interface

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(i.e., Ovid, EBSCO…): [mandatory]
OVID

Research Question
(Describe the purpose of the search) [mandatory]
What do healthcare professionals know about conscience?
How do healthcare professionals make meaning of conscience?
What interventions have been done to support healthcare professionals to understand what conscience is and to use it in healthcare education and practice?
How are healthcare professionals supported to use their conscience in medicine, nursing and midwifery?

PICO Format
(Outline the PICOs for your question — i.e., Patient, Intervention, Comparison, Outcome, and Study Design — as applicable)
P - Healthcare professionals, specifically physicians, nurses, and midwives
I - Interventions (general), education, training, mentoring, quality improvement, tools, screening
C - NA
O - Conscience, empirical ethics
S - NA

Inclusion Criteria
(List criteria such as age groups, study designs, etc., to be included) [optional]
English, scholarly, peer-reviewed, full text articles from 2000 onward pertaining to interventions on conscience for healthcare professionals (population). Interventions include experimental, quasi-experimental and non- experimental studies consisting of clinical trials, qualitative, quantitative and mixed-method studies. Unpublished articles will not be included.

Exclusion Criteria
(List criteria such as study designs, date limits, etc., to be excluded) [optional]
Articles that solely focus on arguing against conscience and issues of conscience for healthcare professionals. Review articles.

Was a search filter applied?
Yes X No

If YES, which one(s) (e.g., Cochrane RCT filter, PubMed Clinical Queries filter)? Provide the source if this is a published filter. [mandatory if YES to previous question — textbox]
Medline Expert Search Filter for Systematic Reviews

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Other notes or comments you feel would be useful for the peer reviewer? [optional]
The PI specifically requested to remove from the search language for ethics and morals following an initial preliminary search in Medline.

Please copy and paste your search strategy here, exactly as run, including the number of hits per line. [mandatory]

Ovid MEDLINE(R) ALL <1946 to April 19, 2021>

1 conscience/ 1573
2 (conscience? or "conscientious* object*").mp. 3695
3 "empirical ethics".mp. 150
4 1 or 2 or 3 3844
5 health personnel/ or exp nurses/ or exp nurse practitioners/ or exp nurse specialists/ or exp nurses, pediatric/ or exp physicians/ 277280
6 (doctor* or physician* or "general practitioner*" or surgeon* or clinician* or nurse* or midwife* or midwives or ((healthcare or "health care" or medical or nursing) adj3 (professional* or personnel))).mp. 1498351
7 5 or 6 1536357
8 (intervention* or train* or educat* or workshop* or seminar* or curricul* or coach* or mentor* or experiment* or tool* or interview* or "focus group" or review* or screening or questionnaire* or survey* or perception* or implication* or barrier* or facilitat* or obstacle* or encourag* or trial* or test* or "quality improvement*").mp. 13896377
9 education/ or curriculum/ or education, distance/ or exp education, professional/ or exp education, continuing/ or exp education, graduate/ or exp education, medical/ or exp education, nursing/ or exp inservice training/ or mentoring/ 367641
10 focus groups/ or interviews as topic/ or "surveys and questionnaires"/ or health care surveys/ or self report/ 614592
11 Quality Improvement/ 27190
12 8 or 9 or 10 or 11 13910462
13 4 and 7 and 12 681
14 meta-analysis.pt. 129734
15 (meta-anal$ or metaanal$).mp. 232007
16 ((quantitativ$ adj3 review$1) or (quantitativ$ adj3 overview$)).mp. 3403
17 ((systematic$ adj3 review$) or (systematic adj3 overview$)).mp. 239417
18 ((methodologic adj3 review$1) or (methodologic adj3 overview$)).mp. 232
19 (integrat$ adj5 research).mp. 10739
20 (quantitativ$ adj3 synthes$).mp. 3146

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21 or/14-20 370682
22 review.pt. or (review$ or overview$).mp. 3928917
23 (medline or medlars or pubmed or index medicus or embase or cochrane).mp. 251818
24 (scisearch or web of science or psycinfo or psychinfo or cinahl or cinhal).mp. 94621
25 (excerpta medica or psychlit or psyclit or current contents or science citation index or sciences citation index or scopus).mp. 35970
26 (hand search$ or manual search$).mp. 12189
27 ((electronic adj3 database$) or (bibliographic adj3 database$) or periodical index$).mp. 43482
28 (pooling or pooled or mantel haenszel).mp. 116964
29 (peto or der simonian or dersimonian or fixed effect$).mp. 19646
30 ((combine$ or combining) adj5 (data or trial or trials or studies or study or result or results)).mp. 133034
31 or/23-30 509089
32 22 and 31 276394
33 21 or 32 468309
34 (hta$ or health technology assessment$ or biomedical technology assessment$).mp. 8409
35 technology assessment, biomedical/ or biomedical technology assessment/ 10225
36 34 or 35 16492
37 33 or 36 482109
38 13 not 37 661
39 limit 38 to (english language and yr="2000 -Current") 431

PEER REVIEW ASSESSMENT: THIS SECTION TO BE FILLED IN BY THE REVIEWER

Reviewer:Liz Dennett
Email:eal@ualbert
Date completed: April 27, 2021

1. TRANSLATION

A ---No revisions
x☐
B --- Revision(s) suggested
☐
C --- Revision(s) required

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☐

If "B" or "C," please provide an explanation or example:

Comment: I think it is appropriate to use a narrow take on the conscience concept here. Of course all exact synonyms should be included and unfortunately I don't have a good enough understanding of the topic to know what is an exact synonym for conscience and what is only related, but I can see the benefit of not being too broad here. If the authors don't use the term conscience in the abstract, the article is unlikely to be relevant.

2. BOOLEAN AND PROXIMITY OPERATORS

A ---No revisions

x☐

B --- Revision(s) suggested

☐

C --- Revision(s) required

☐

If "B" or "C," please provide an explanation or example:

3. SUBJECT HEADINGS

A ---No revisions

x☐

B --- Revision(s) suggested

☐

C --- Revision(s) required

☐

If "B" or "C," please provide an explanation or example:

4. TEXT WORD SEARCHING

A ---No revisions

☐

B --- Revision(s)suggested

X☐

C --- Revision(s) required

☐

If "B" or "C," please provide an explanation or example:

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Line 6: I would add health to the first set of brackets and provider* or practitioner* to the second. Terms such as health practitioner* or healthcare provider* can pick up physician / nurse focused articles.

Also curious why all the physician specialties were removed from the preliminary search although I assume there was a good reason for this.

The intervention concept is tough. You have done a good job coming up with terms that will help you find the publication types and study designs that the protocol indicates you are looking for. I just worry that there might be additional types of studies that are not well described that may be relevant.

Here are just some suggestions of terms that might be relevant that you COULD consider (but by no means have to) What about practice guidelines as an intervention? Would that make sense here? Or Dagmara has this list of KT terms. Would any of them be relevant?: https://docs.google.com/document/d/1v5Hyfpc-BICRmfNOaqp06H5xYLVvW0Ns5RDM2MwbsiA/edit?usp=sharing . Additional QI suggestions: (QI adj6 (intervention* or initiative* or strateg* or program* or campaign* or implement*)).mp. (Practice or behavio* or organizational) adj3 chang*).mp. patient safety.mp. Quality assurance.mp.

Additional barrier / experiential terms: Impediment* or impede* or challenges or obstruct* or hurdle* OR Experience* or perceive* or perspective* or self-report*

5. SPELLING, SYNTAX, AND LINE NUMBERS

A ---No revisions

X☐

B --- Revision(s)suggested

☐

C --- Revision(s) required

☐

If "B" or "C," please provide an explanation or example:

6. LIMITS AND FILTERS

A ---No revisions

☐

B --- Revision(s) suggested

x☐

C --- Revision(s) required

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☐

If "B" or "C," please provide an explanation or example:
Highly sensitive filters are good for finding those study types but to exclude a publication type, it is safer to use a highly precise filter – such as (systematic* review or meta-anal* or metanal* or hta or health technology assessment).ti. or systematic review/ or meta-analysis.pt. or technology assessment, biomedical/ or biomedical technology assessment/ or review.pt. This may not make a big difference to your numbers but I still think it is better to take this approach

OVERALL EVALUATION (NOTE: If one or more "revision required" is noted above, the response below must be "revisions required")

A ---No revisions
☐
B --- Revision(s) suggested
x☐
C --- Revision(s) required
☐

Additional comments: none.

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PRISMA-P 2015 Checklist

This checklist has been adapted for use with protocol submissions to *Systematic Reviews* from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	x		Title page, pp 7
Update	1b	If the protocol is for an update of a previous systematic review, identify as such		x	NA
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract		X	
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	x		Title page, manuscript online submission process
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	x		Pp 15
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments		x	NA
Support					
Sources	5a	Indicate sources of financial or other support for the review	x		Pp 15
Sponsor	5b	Provide name for the review funder and/or sponsor	x		Pp 15
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol		x	NA
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	x		Pp 4-6

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	x		Pp 6-8
METHODS					
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	x		Pp 7-9
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	x		Pp 7-9
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	x		Table 1
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	x		Pp 11-13
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	x		Pp 11-13
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	x		Pp 11-13
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	x		Pp 11-13
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	x		Pp 8,13,14
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	x		Pp 13
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	x		Pp 13,14
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)		x	NA
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	x		Pp 13,14
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	x		Pp 13,14

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)		x	NA
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	x		Pp 14

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BMJ Open

Effectiveness of interventions on healthcare professionals' understanding and use of conscience: A systematic review protocol.

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-053880.R1
Article Type:	Protocol
Date Submitted by the Author:	28-Dec-2021
Complete List of Authors:	Lamb, Christina; University of Alberta, Faculty of Nursing Kennedy, Megan; University of Alberta, John W. Scott Health Sciences Library Clark, Alex; Athabasca University, Faculty of Health Disciplines Pituskin, Edith; University of Alberta, Faculty of Nursing Kirkwood , Ken ; Western University, Health Sciences Babenko-Mould , Yolanda ; Western University, Arthur Labatt Family School of Nursing
Primary Subject Heading:	Ethics
Secondary Subject Heading:	Global health
Keywords:	ETHICS (see Medical Ethics), Health policy < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, EDUCATION & TRAINING (see Medical Education & Training)

SCHOLARONE™
Manuscripts

Title: Effectiveness of interventions on healthcare professionals' understanding and use of conscience: A systematic review protocol.

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Keywords: Conscience, healthcare professionals, interventions, systematic review protocol.

Word count: 3291

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Abstract

Introduction: Conscience is central to moral decision making. In the context of morally pluralistic workplaces today, healthcare professionals' conscience may prompt them to make moral decisions to refrain from providing services they morally disagree with. However, such decisions are largely viewed as contentious, giving rise to polarizing arguments for and against healthcare professionals' freedom of conscience. Yet, little work has been done to understand and support healthcare professionals' conscience. Instead, the rising polarity related to healthcare professionals' freedom of conscience stems from a central lack of understanding of what conscience is and the relevance it holds for healthcare professionals' clinical practice. Therefore, the degree and extent to which healthcare professionals are supported to understand and use their conscience is unknown. The objective of this review is to critically analyze the scholarly evidence available to ascertain the effectiveness of interventions that support healthcare professionals to understand and use their conscience in care practice.

Methods and Analyses: At least two reviewers will systematically review ten inter-disciplinary, scholarly databases to examine qualitative, quantitative and mixed-methods studies including clinical trials pertaining to interventions related to conscience for healthcare professionals. Databases to be searched include: the Cochrane Controlled Register of Trials, Medline, EMBASE, PsycINFO, CINAHL, Academic Search Complete, ATLA Religion Database, Religion and Philosophy Collection, PhilPapers and Scopus. Databases were searched in May, 2021. Study screening, selection, extraction and risk of bias assessments on each study using the Mixed Methods Appraisal Tool will be independently conducted by independent reviewers. Descriptive data synthesis will be carried out. Statistical analysis and meta-analysis will be conducted as relevant, based on homogeneity of findings. The quality of the aggregate evidence will be assessed using the Grading of Recommendations Assessment, Development and Evaluation Tool criteria.

Ethics and dissemination: Ethical approval is not required for this review. This protocol will not involve individual patient information endangering participant rights. The results will be reported in a peer-reviewed journal and disseminated at conferences.

PROSPERO registration number: CRD42021256943

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Strengths and limitations (5 bullet point sentences, related to the methods)

- This study aims to examine interventions used to enhance HCPs' understanding and use of conscience.
- Study selection, data extraction and risk of bias assessments will be done independently by two-four reviewers.
- Studies included in this review will be limited to English language articles, which may lead to a risk for publication bias.
- The strength of evidence in this study will be evaluated according to the Grading of Recommendations Assessment, Development and Evaluation criteria.
- Inclusion of mixed-method interventions may not allow for statistical analysis of findings which might lead to inconsistent reporting of outcomes.

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Introduction

Conscience is a fundamental human freedom and an essential concept for bioethics and the health sciences.[1,2,3] Conscience is the ability to make decisions based on information derived from moral knowledge in order to choose to do something that is ethical or refrain from doing something unethical.[4,5] As such, conscience is central to ethical healthcare in which healthcare professionals (HCPs) are regularly required to make moral decisions to engage in practice that is ethical and beneficial for their patients.

However, in bioethics and the health sciences today, little attention has been paid to the meaning of conscience as it pertains to healthcare professionals' (HCP) moral decision making. Instead, issues around conscience, such as conscientious objection, are becoming increasingly polarized in today's morally pluralistic societies and healthcare communities owing to the lack of agreement that may occur in pluralistic contexts.[6-8] Subsequently, disagreement on how HCPs express their issues of conscience is also increasing owing to the lack of understanding regarding what conscience is in the first place.[6] This knowledge gap is problematic because the scholarly literature is fraught with academic dispute over how to handle HCP's issues of conscience.[6,-8]Amidst this debate, little attention is being given to HCPs with respect to expressing their issues of conscience and what conscience and issues of conscience mean to them. As an initial effort to address this gap, a systematic review is needed to assess the state of the literature on how HCPS are being supported to both understand and use their conscience. Namely, identifying and assessing what interventions exist to do so and how effective they are.

While many HCP groups exist today, for the purpose of focus of scope and feasibility for this review, HCPs will consist of physicians, nurses and midwives who have all been

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documented in the scholarly literature as having to make ethical decisions in the course of their practice which involves the use of their conscience.[9-11] Additionally, international organizations such as the World Medical Association, the International Council of Nurses and the International Confederation of Midwives stipulate that conscience and the human right to freedom of conscience are relevant to medicine, nursing and midwifery.[2,3,12] It is anticipated that this initial review will inform subsequent reviews that will involve other HCP groups.

The research that has been done on the meaning of conscience for HCPs indicates that physicians, nurses and midwives alike voice that conscience is important to their ability to make moral meaning of their ethical practice.[9,10,13] Importantly, conscience is well taken up in the philosophical and theological literature. Within this scholarly space, conscience has a long-standing conceptual history.[13, 14, 1] In this body of literature, conscience has emerged as a necessary component for being a moral person.[15,16] More recently, empirical work in moral psychology is revealing compelling empirical evidence that morality is central to human behaviour.[17-19] Philosophy and theology are also taken up in bioethics, which is an inter-disciplinary discipline devoted to asking essential life questions that support healthcare providers and patients alike to make ethical decisions in the context of healthcare.[20,21] Owing to the traction that conscience has in these inter-disciplinary disciplines, we will use an inter-disciplinary approach to answer our research questions: 1) what interventions exist to support HCPs' use and understanding of conscience in bioethics, clinical education and practice? ; 2) how effective are these interventions? ; 3) is there an intervention that draws on an inter-disciplinary approach to conscience to inform and support these interventions?; 4) how effective did HCPs find these interventions?

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Supporting HCPs to use their conscience and increasing an awareness of the significance that conscience holds for ethical decision making might start to address the gap related to understanding the relevance of conscience in bioethics and healthcare communities today. It may also offer a morally inclusive way to support HCPs' moral decision making in practice. Advancing more understanding in relation to conscience and the role it holds in the human experience across healthcare contexts has the potential for global impact given that freedom of conscience is a universal, human right. This review might generate further empirical ethical research to benefit HCPs' ethical practice and the patient populations they care for.

To start to address this knowledge gap related to HCPs' understanding and use of conscience, we will conduct a systematic review (SR) of the interdisciplinary, scholarly literature about interventions utilized to support HCPs' understanding and use of conscience. To the best of our knowledge, no systematic reviews have been undertaken in relation to this gap. Disciplines that will be considered as previously mentioned are: nursing; medicine; midwifery; bioethics; philosophy; theology; religious studies and moral psychology.

To answer the research questions the primary objectives of this review study are:

1. To review the evidence about the interventions that were utilized.
2. To determine whether the interventions were effective in providing or improving HCPs understanding and use of conscience.
3. To explore the effectiveness of these interventions as they pertain to the physician, nursing and midwife subgroups and to assess if there is any difference in the effectiveness of the interventions amongst them.

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Methods

Research design and methodology

We will conduct a SR of the published, scholarly, inter-disciplinary literature to examine and explore the effectiveness of interventions utilized to support HCP's understanding and use of conscience in healthcare practice. To initiate the SR study we will follow this review protocol which will be registered in PROSPERO (CRD42021256943). The review will be reported in adherence to the Preferred Reporting Items for Systematic Review and Meta-analysis 2020 Statement for standardized reporting guidelines, the PRISMA extension checklist for searches (PRISMA-S) and the PRISMA 2020 statement for writing abstracts.[22-24] The PRISMA-S checklist will be used to report and document the literature search.[23] This protocol was generated following the Preferred Reporting Item for Systematic review and Meta-Analysis Protocols (PRISMA-P) checklist.[24] (See Supplementary File 1).

Eligibility criteria

Inclusion criteria

To isolate eligible studies, we will follow the Population, Intervention, Comparators, Outcomes and Study design (PICOS) framework as stipulated in the PRISMA (2020) statement as follows:

Population: Healthcare professionals who engage in clinical practice across any practice setting as identified across the sub-groups of physicians, nurses, and midwives. **Interventions:** Interventions that will be examined are those that aim to improve HCPs' use and understanding of conscience. Possible interventions are, but not limited to: experiments; education and quality improvement initiatives; interventions by leadership teams to enhance and foster HCPs' use and

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understanding of conscience; interviews; focus groups; policies, regulations and practices aimed at changing how HCPs use their conscience in practice settings; experiential inquiries and narrative studies as well as studies that include trials and pre-post-test data sets to examine how effective the interventions are. *Sample interventions:* studies and papers to be included are those that use primary data and aim to support, enhance, show and address the use and understanding and use of conscience for HCPs. This may occur through terms closely related to conscience such as conscientious objection. For instance, studies involving the Stress of Conscience Questionnaire and the Perception of Conscience Questionnaire as part of an intervention will be included since they directly relate to the research questions of this SR.[25, 26]

Comparators: studies that included controls or comparators of interventions. **Outcomes:** the types of outcomes that will be sought from the data are the: (1) findings that identify what interventions exist that aim to support HCPs to understand and use their conscience; (2) reports on how HCPs felt supported to use and / or understand their conscience; (3) effects of HCPs being able to voice their issues of conscience in healthcare practice across the sub-groups; (4) what effects that support for conscience has on HCPs across these sub-groups in relation to the quality of care they provide to patients, in relation to themselves, working with colleagues and in their professions and 5) how effective these interventions were. **Study designs:** studies involving interventions may include experimental, quasi-experimental and non-experimental studies consisting of clinical trials, qualitative, quantitative and mixed-method studies.

Exclusion criteria

For this initial protocol review articles pre-dating 2000 are ineligible owing to the limited empirical research about interventions specifically related to HCPs' use and understanding of

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conscience prior to this date. Unpublished articles including theses and dissertations will not be included. Papers that utilize secondary data sets such as protocols and reviews are also ineligible.

Search Strategy:

This review will be reported in adherence to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement with methodological guidance from the Cochrane Handbook of Systematic Reviews of Interventions.[22,27] Further, a systematic search will be developed and conducted by an experienced health sciences librarian (MK), reported in adherence to the PRISMA for Searching (PRISMA-S) extension.[23]

A preliminary search using the database, Medline, was conducted in April 2021 by MK to determine the feasibility of the review and inform the systematic search strategy (see Table 1 for full search strategy). The search strategy was derived from three main concepts: 1) conscience; 2) healthcare professionals, specifically physicians (including specialists), nurses, and midwives; 3) interventions, assessment, and quality improvement. Systematic, scoping, and other types of synthesis reviews and their protocols were removed from the results. Date limitations will be removed, and systematic reviews will be included in the updated search for the full-scale review.

(Insert Table 1 here:)

Table 1: **Search Strategy for Medline(R) ALL via OVID (1946 - April 26, 2021).**
Date of search: 27 April 2021

1	conscience/	1574
2	(conscience? or "conscientious* object*").mp.	3707
3	"empirical ethics".mp.	151
4	1 or 2 or 3	3857
5	health personnel/ or exp nurses/ or exp nurse practitioners/ or exp nurse specialists/ or exp nurses, pediatric/ or exp physicians/	278364
6	(doctor* or physician* or "general practitioner*" or GP or surgeon* or anesthesiologist* or cardiologist* or dermatologist* or radiologist* or internist* or geneticist* or neurologist* or obstetrician* or gyn?ecologist* OBGYN or OB-GYN or ophthalmologist*	

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or pathologist* or p?ediatrician* or psychiatrist* or oncologist* or urologist* or allergist* or endocrinologist* or gastroenterologist* or geriatrician* or hospitalist* or nephrologist* or pulmonologist* or rheumatologist* or clinician* or nurse* or midwife* or midwives or ((healthcare or "health care" or health or medical or nursing) adj3 (professional* or personnel or practitioner* or provider*))).mp. 1905235

7 5 or 6 1907349

8 (intervention* or train* or educat* or workshop* or seminar* or curricul* or coach* or mentor* or experiment* or tool* or interview* or "focus group" or review* or screening or questionnaire* or survey* or perception* or implication* or barrier* or facilitat* or obstacle* or encourag* or impediment* or impede* or challeng* or obstruct* or hurdle or experience* or perceive* or perspective* or perception* or "self-report" or trial* or test* or "quality improvement*" or "quality assurance" or "patient safety").mp. 14879880

9 ((QI or "quality improvement*") adj6 (intervention* or initiative* or strateg* or program* or campaign* or implement*)).mp. 18223

10 ((Practice or behavio* or organizational) adj3 chang*).mp. 88479

11 education/ or curriculum/ or education, distance/ or exp education, professional/ or exp education, continuing/ or exp education, graduate/ or exp education, medical/ or exp education, nursing/ or exp inservice training/ or mentoring/ 368239

12 focus groups/ or interviews as topic/ or "surveys and questionnaires"/ or health care surveys/ or self report/ 617218

13 Quality Improvement/27360

14 (((research or knowledge or evidence) adj2 (uptake or "use" or diffus$ or disseminat$ or utiliz$ or utilis$ or transfer$ or translat$ or implement$ or adopt$)) or (innovation$ adj2 adopt$) or (innovation$ adj2 disseminat$) or "research into practice" or "evidence into practice" or "knowledge to action" or "know do gap" or (knowledge adj (mobilization or exchange)) or "translational science" or "implementation science").ti,ab. 70172

15 exp "Diffusion of Innovation"/ or Organizational Innovation/ 43888

16 (evidence-informed adj (healthcare or health care or decision making)).ti,ab. 241

17 (guideline* adj2 (introduc*or issu* or impact or effect* or distribut* or adher* or compl* or utiliz* or utilis* or "use" or uptake or diffuse* or transfer* or implement* or translat* or disseminat* or adopt*)).ti,ab. 20497

18 exp *Evidence-Based Practice/ or Information dissemination/ or Knowledge management/ or Health Knowledge, Attitudes, Practice/ 166249

19 or/8-1814952921

20 4 and 7 and 19893

21 ("systematic* review" or "scoping review" or "synthesis review" or "realist review" or "integrative review" or "meta-anal*" or "metanal* meta-synths?s" or metasynthes?s or meta-narrative or metanarrative or "umbrella review" or hta or "health technology assessment").ti. or systematic review/ or meta-analysis.pt. or technology assessment, biomedical/ or biomedical technology assessment/ or review.pt. 2950266

22 20 not 21 762

23 limit 22 to (english language and yr="2000 -Current") 510

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In order to enhance the quality of the systematic search, the preliminary search strategy for Medline was evaluated by an external health sciences librarian using Peer Review of Electronic Search Strategies (PRESS) guidelines.[28] No major revisions were required following the PRESS evaluation but the following recommendations were made: 1) strategy for capturing synthesis reviews was refined to make it more precise for exclusion purposes; 2) additional terminology related to knowledge translation were added to the intervention concept (see Supplementary File 2). The search strategy was revised to include these recommendations and in total, 510 articles were identified from the preliminary search. Out of these results, 30 were deemed relevant for inclusion and confirmed the feasibility of this review.

To identify all relevant published studies, the following databases will be searched:

- Medline via OVID (1946 – Present)
- EMBASE via OVID (1974 – Present)
- PsycINFO via OVID (1806 – Present)
- Cumulative Index for Nursing and Allied Health Literature (CINAHL) via EBSCOhost (1936 – Present)
- Academic Search Complete via EBSCOhost (1887 – Present)
- ATLA Religion Database via EBSCOhost (13th C. – Present)
- Religion and Philosophy Collection via EBSCOhost (1911 – Present)
- PhilPapers via Philosophical Research Online (2001 – Present)
- Scopus via Elsevier (1976 – Present)
- Cochrane Controlled Register of Trials (CENTRAL) via Wiley (1993 – Present)

These databases will be searched using a combination of natural language vocabulary (keywords) and controlled terms (subject headings) wherever they are available, based on the

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previously identified search concepts. Records identified through database searches will be exported in complete batches and duplicates will automatically be removed upon import to the systematic review management software, Covidence.[29] An update of the searches is planned for six months after search results are exported or one month prior to manuscript submission for publication, whichever is sooner. There is no patient or public involvement in this study.

Screening

Articles will be screened by at least two, independent related to the eligibility criteria. Prior to screening, the reviewers will generate screening questions related to both title and abstract and full screening processes which will be done through Covidence software. To facilitate the screening process, a minimum of two reviewers will independently screen all records by title and abstract and retain articles that meet the eligibility and screening question criteria. Included articles for full-text review will again be screened in full by a minimum of two, independent reviewers. Prior to each step in the screening process, reviewers will initially screen approximately ten records and meet to discuss the process to flesh out any questions or disagreements and before carrying out the full screening process. Reviewers will meet throughout each screening process to discuss progress and any conflicts that arise. Throughout the screening and data extraction processes the PI will be responsible for resolving disagreements if consensus is not reached amongst the researchers. Full articles that meet the eligibility criteria will be extracted for data corresponding to variables pre-determined by the research team. The screening and selection process will be shown in a PRISMA flow chart.

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Data extraction

Data from included, full text articles will be extracted by a minimum of two, independent reviewers according to pre-set headings developed by the research team. Data headings will be comprised of the following variables in relation to the research questions, PICOS criteria as well as primary objectives: full reference of the articles; region and country the study was conducted in; research objectives; study design and methods (sample size, comparison types); population under study including sub-groups; description of the intervention and control interventions (where applicable); effects of the intervention (whether the study supported HCPs to understand and use their conscience and how, as well as to what extent) in relation to the study outcomes listed in the PICOS criteria which will be captured as significant findings in terms of outcomes, effects of outcomes and limitations. Extracted data will be entered into Word or Excel documents under the data extraction headings to be tabulated for ease of data synthesis and reporting.

Critical appraisal: Assessing risk of bias

Since mixed-methods studies will be included in this SR, extracted data will be critically appraised for methodological quality using the Mixed Methods Assessment Tool (MMAT).[30] At least two, independent researchers will perform this assessment in each individual study. The MMAT is a critical appraisal tool aimed at assessing the quality of studies included in systematic reviews. All studies are appraised against 5 categories of criteria for validity and rigour in relation to study designs comprised of either qualitative, quantitative (randomized trials, non-randomized and descriptive studies) and mixed-methods papers. Reviewers capture their assessments according to core criteria across study designs by either responding ‘yes’, ‘no’ or

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'can't tell'. Two, independent reviewers will appraise each study using the MMAT and resolve any discrepancies with a third reviewer or the PI. Assessments for each study will be tabulated and presented for data synthesis.

Data synthesis

It is expected that the studies identified will have high methodological heterogeneity – featuring a mix of qualitative, quantitative and mixed-method studies. Accordingly, extracted data will be synthesized descriptively according to relevance of interventions by effect (whether they aimed to support HCPs in understanding and /or using their conscience). The data will be discussed in text and tables to summarize and describe the characteristics and relationship of findings across the included studies.[22] Findings will be reported according to fixed categories including: the study year, purpose, design and setting, population, intervention and effect, primary outcome and limitations. Since this SR will include non-quantitative studies, the heterogeneity of the findings will determine whether there is sufficient data to conduct statistical meta-analysis via pooling of outcomes data from sufficiently similar intervention studies. If this is possible, given the wide likely variability across studies, a random effects model will be used for the meta-analysis. If statistical analysis is not applicable, all data will be reported through a narrative descriptive analysis. Quality of the included studies will be assessed using the appropriate checklist tools from the Critical Appraisal Skills Program (CASP); with mixed method studies assessed using the Mixed Method Appraisal Tool (MMAT) for mixed-method studies.[31, 30]

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Confidence in cumulative evidence assessment

The Grading of Recommendations Assessment, Development and Evaluation (GRADE) tool will be used to evaluate the strength of the evidence of the included studies.[32]

Quality Assurance

The proposed SR study will follow the PRISMA (2020) format for reporting systematic reviews.

Discussion

This study will review current, inter-disciplinary research studies explicating interventions that were used to understand and / or support HCP's use of conscience. The results of this systematic review will be useful in identifying what concrete supports exist in relation to moral inclusion of HCP's freedom of conscience. Additional gaps may be identified leading to further empirical ethics research necessary to support HCP populations to navigate today's morally complex healthcare context.

Acknowledgements: We thank the second research librarian for their PRESS review of the initial Medline search. The PI thanks her colleague, Dr. Matthias Hoben, for his review of the initial manuscript.

Ethics and dissemination: Ethics approval is not required for this study. Findings will be disseminated through publication in a peer-reviewed journal and pertinent conferences.

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Author's contributions: CL conceptualized, drafted and revised the protocol. MK and CL created the search strategy. MK conducted the initial Medline search, had the search strategy peer-reviewed, and searched the databases. AMC provided statistical support. AMC, EP and KK contributed to the development of the search criteria and conceptualization of the protocol. KK contributed to the assessment of ROB tools. YBM contributed to the assessment of the strengths and limitations of the research methods. All authors read, provided feedback or approved the manuscript for submission.

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Disclaimer: The work undertaken in this research and all the content and conclusions therein are that of the authors and not the funders.

Competing interests' statement: None declared.

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PRESS Guideline — Search Submission & Peer Review Assessment

SEARCH SUBMISSION: THIS SECTION TO BE FILLED IN BY THE SEARCHER

Searcher: MK
Email: Will be provided after peer review for the purpose of anonymous peer-review of the protocol manuscript.
Date submitted: April 20, 2021
Date requested by: TBD.
[Maximum = 5 working days]

Systematic Review Title:
Effectiveness of interventions on healthcare professionals' understanding and use of conscience: A systematic review protocol.

This search strategy is …

X	My PRIMARY (core) database strategy — First time submitting a strategy for search question and database
	My PRIMARY (core) strategy — Follow-up review NOT the first time submitting a strategy for search question and database. If this is a response to peer review, itemize the changes made to the review suggestions
	SECONDARY search strategy— First time submitting a strategy for search question and database
	SECONDARY search strategy — NOT the first time submitting a strategy for search question and database. If this is a response to peer review, itemize the changes made to the review suggestions

Database
(i.e., MEDLINE,CINAHL…): [mandatory]
MEDLINE

Interface

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(i.e., Ovid, EBSCO…): [mandatory]
OVID

Research Question
(Describe the purpose of the search) [mandatory]
What do healthcare professionals know about conscience?
How do healthcare professionals make meaning of conscience?
What interventions have been done to support healthcare professionals to understand what conscience is and to use it in healthcare education and practice?
How are healthcare professionals supported to use their conscience in medicine, nursing and midwifery?

PICO Format
(Outline the PICOs for your question — i.e., Patient, Intervention, Comparison, Outcome, and Study Design — as applicable)
P - Healthcare professionals, specifically physicians, nurses, and midwives
I - Interventions (general), education, training, mentoring, quality improvement, tools, screening
C - NA
O - Conscience, empirical ethics
S - NA

Inclusion Criteria
(List criteria such as age groups, study designs, etc., to be included) [optional]
English, scholarly, peer-reviewed, full text articles from 2000 onward pertaining to interventions on conscience for healthcare professionals (population). Interventions include experimental, quasi-experimental and non- experimental studies consisting of clinical trials, qualitative, quantitative and mixed-method studies. Unpublished articles will not be included.

Exclusion Criteria
(List criteria such as study designs, date limits, etc., to be excluded) [optional]
Articles that solely focus on arguing against conscience and issues of conscience for healthcare professionals. Review articles.

Was a search filter applied?
Yes X No

If YES, which one(s) (e.g., Cochrane RCT filter, PubMed Clinical Queries filter)? Provide the source if this is a published filter. [mandatory if YES to previous question — textbox]
Medline Expert Search Filter for Systematic Reviews

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Other notes or comments you feel would be useful for the peer reviewer? [optional]
The PI specifically requested to remove from the search language for ethics and morals following an initial preliminary search in Medline.

Please copy and paste your search strategy here, exactly as run, including the number of hits per line. [mandatory]

Ovid MEDLINE(R) ALL <1946 to April 19, 2021>

1 conscience/ 1573
2 (conscience? or "conscientious* object*").mp. 3695
3 "empirical ethics".mp. 150
4 1 or 2 or 3 3844
5 health personnel/ or exp nurses/ or exp nurse practitioners/ or exp nurse specialists/ or exp nurses, pediatric/ or exp physicians/ 277280
6 (doctor* or physician* or "general practitioner*" or surgeon* or clinician* or nurse* or midwife* or midwives or ((healthcare or "health care" or medical or nursing) adj3 (professional* or personnel))).mp. 1498351
7 5 or 6 1536357
8 (intervention* or train* or educat* or workshop* or seminar* or curricul* or coach* or mentor* or experiment* or tool* or interview* or "focus group" or review* or screening or questionnaire* or survey* or perception* or implication* or barrier* or facilitat* or obstacle* or encourag* or trial* or test* or "quality improvement*").mp. 13896377
9 education/ or curriculum/ or education, distance/ or exp education, professional/ or exp education, continuing/ or exp education, graduate/ or exp education, medical/ or exp education, nursing/ or exp inservice training/ or mentoring/ 367641
10 focus groups/ or interviews as topic/ or "surveys and questionnaires"/ or health care surveys/ or self report/ 614592
11 Quality Improvement/ 27190
12 8 or 9 or 10 or 11 13910462
13 4 and 7 and 12 681
14 meta-analysis.pt. 129734
15 (meta-anal$ or metaanal$).mp. 232007
16 ((quantitativ$ adj3 review$1) or (quantitativ$ adj3 overview$)).mp. 3403
17 ((systematic$ adj3 review$) or (systematic adj3 overview$)).mp. 239417
18 ((methodologic adj3 review$1) or (methodologic adj3 overview$)).mp. 232
19 (integrat$ adj5 research).mp. 10739
20 (quantitativ$ adj3 synthes$).mp. 3146

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21 or/14-20 370682
22 review.pt. or (review$ or overview$).mp. 3928917
23 (medline or medlars or pubmed or index medicus or embase or cochrane).mp. 251818
24 (scisearch or web of science or psycinfo or psychinfo or cinahl or cinhal).mp. 94621
25 (excerpta medica or psychlit or psyclit or current contents or science citation index or sciences citation index or scopus).mp. 35970
26 (hand search$ or manual search$).mp. 12189
27 ((electronic adj3 database$) or (bibliographic adj3 database$) or periodical index$).mp. 43482
28 (pooling or pooled or mantel haenszel).mp. 116964
29 (peto or der simonian or dersimonian or fixed effect$).mp. 19646
30 ((combine$ or combining) adj5 (data or trial or trials or studies or study or result or results)).mp. 133034
31 or/23-30 509089
32 22 and 31 276394
33 21 or 32 468309
34 (hta$ or health technology assessment$ or biomedical technology assessment$).mp. 8409
35 technology assessment, biomedical/ or biomedical technology assessment/ 10225
36 34 or 35 16492
37 33 or 36 482109
38 13 not 37 661
39 limit 38 to (english language and yr="2000 -Current") 431

PEER REVIEW ASSESSMENT: THIS SECTION TO BE FILLED IN BY THE REVIEWER

Reviewer:Liz Dennett
Email:eal@ualbert
Date completed: April 27, 2021

1. TRANSLATION

A ---No revisions
x☐
B --- Revision(s) suggested
☐
C --- Revision(s) required

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☐

If "B" or "C," please provide an explanation or example:

Comment: I think it is appropriate to use a narrow take on the conscience concept here. Of course all exact synonyms should be included and unfortunately I don't have a good enough understanding of the topic to know what is an exact synonym for conscience and what is only related, but I can see the benefit of not being too broad here. If the authors don't use the term conscience in the abstract, the article is unlikely to be relevant.

2. BOOLEAN AND PROXIMITY OPERATORS

A ---No revisions

x☐

B --- Revision(s) suggested

☐

C --- Revision(s) required

☐

If "B" or "C," please provide an explanation or example:

3. SUBJECT HEADINGS

A ---No revisions

x☐

B --- Revision(s) suggested

☐

C --- Revision(s) required

☐

If "B" or "C," please provide an explanation or example:

4. TEXT WORD SEARCHING

A ---No revisions

☐

B --- Revision(s)suggested

X☐

C --- Revision(s) required

☐

If "B" or "C," please provide an explanation or example:

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Line 6: I would add health to the first set of brackets and provider* or practitioner* to the second. Terms such as health practitioner* or healthcare provider* can pick up physician / nurse focused articles.

Also curious why all the physician specialties were removed from the preliminary search although I assume there was a good reason for this.

The intervention concept is tough. You have done a good job coming up with terms that will help you find the publication types and study designs that the protocol indicates you are looking for. I just worry that there might be additional types of studies that are not well described that may be relevant.

Here are just some suggestions of terms that might be relevant that you COULD consider (but by no means have to) What about practice guidelines as an intervention? Would that make sense here? Or Dagmara has this list of KT terms. Would any of them be relevant?: https://docs.google.com/document/d/1v5Hyfpc-BICRmfNOaqp06H5xYLVvW0Ns5RDM2MwbsiA/edit?usp=sharing . Additional QI suggestions: (QI adj6 (intervention* or initiative* or strateg* or program* or campaign* or implement*)).mp. (Practice or behavio* or organizational) adj3 chang*).mp. patient safety.mp. Quality assurance.mp.

Additional barrier / experiential terms: Impediment* or impede* or challenges or obstruct* or hurdle* OR Experience* or perceive* or perspective* or self-report*

5. SPELLING, SYNTAX, AND LINE NUMBERS

A ---No revisions

X☐

B --- Revision(s)suggested

☐

C --- Revision(s) required

☐

If "B" or "C," please provide an explanation or example:

6. LIMITS AND FILTERS

A ---No revisions

☐

B --- Revision(s) suggested

x☐

C --- Revision(s) required

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☐

If "B" or "C," please provide an explanation or example:
Highly sensitive filters are good for finding those study types but to exclude a publication type, it is safer to use a highly precise filter – such as (systematic* review or meta-anal* or metanal* or hta or health technology assessment).ti. or systematic review/ or meta-analysis.pt. or technology assessment, biomedical/ or biomedical technology assessment/ or review.pt. This may not make a big difference to your numbers but I still think it is better to take this approach

OVERALL EVALUATION (NOTE: If one or more "revision required" is noted above, the response below must be "revisions required")

A ---No revisions
☐
B --- Revision(s) suggested
x☐
C --- Revision(s) required
☐

Additional comments: none.

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PRISMA-P 2015 Checklist

This checklist has been adapted for use with protocol submissions to *Systematic Reviews* from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	x		Title page
Update	1b	If the protocol is for an update of a previous systematic review, identify as such		x	NA
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	x		Abstract and pp 7
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	x		Title page
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	x		Pp 16
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments		x	NA
Support					
Sources	5a	Indicate sources of financial or other support for the review	x		Pp 16
Sponsor	5b	Provide name for the review funder and/or sponsor		x	NA
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol		x	NA
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	x		Pp 4-6
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	x		Pp 7-8
METHODS					

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	x		Pp 7-9
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	x		Pp 9-12
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	x		Table 1
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	x		Pp 12,14
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	x		Pp 12-14
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	x		Pp 13
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	x		Pp 13
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	x		Pp 7,8,13
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	x		Pp 13,14
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	x		Pp 14
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)		x	NA
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	x		Pp 14,15
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	x		Pp 14
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)		x	NA
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	x		Pp 15

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